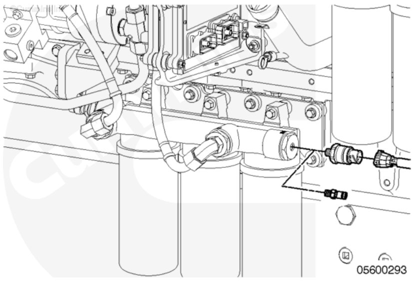
Trainings

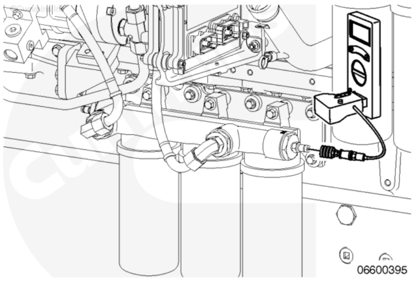
Initial Check

Note : Shown is the Stage 2 filter head for the QSK38 engine. Although different in appearance, the procedure remains the same for the QSK engines.

Remove the pressure sensor and replace it with a Compuchek™ fitting,

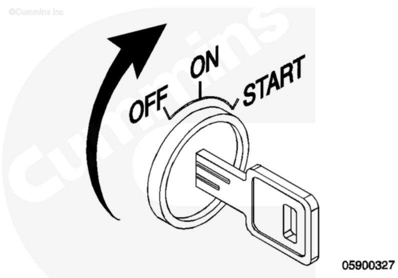


Connect the pressure adapter, Part Number 3164491, or equivalent, and the digital multimeter, Part Number 3164488, 3164489, or equivalent, to the Compuchek™ fitting.



The fuel lift pump will operate for 120 seconds when the key is switched to the ON position. The fuel lift pump will also operate while the engine is cranking until the fuel rail pressure reaches 200 bar [2900 psi].

At the end of 120 seconds of operation the fuel lift pump **must** produce pressure. If the lift pump does **not**, then the lift pump **must** be replaced.



Minimum Fuel Lift Pressure		
kpa		psi
300	MIN	43.5

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

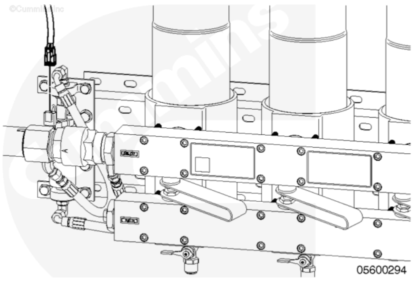
- Disconnect the battery cables. See equipment manufacturer service information.
- Close the fuel supply valve.

Remove

Note : Applications with lift pumps mounted on Stage 1 filter heads are explained in the following steps.

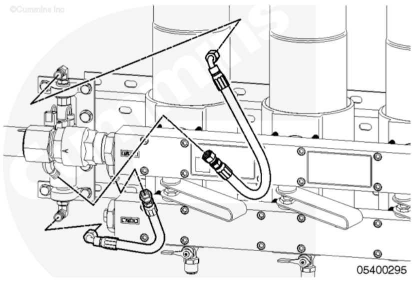
Note : Shown is the Stage 2 filter head for QSK38 series engines. Although different in appearance, the procedure remains the same.

Disconnect the electrical connections to the fuel lift pump.



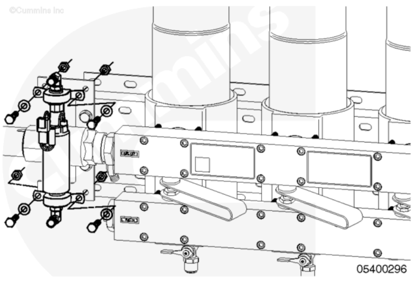
⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting to performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, arcing equipment, or welding equipment) in the work area.

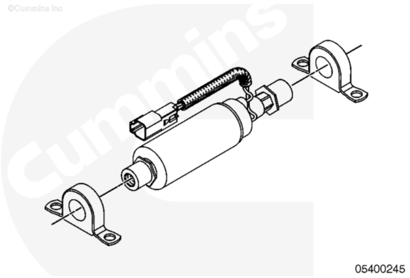


Disconnect the supply and discharge flexible hoses from the lift pump.

Remove the four capscrews securing the fuel lift pump to the fuel filter mounting bracket.

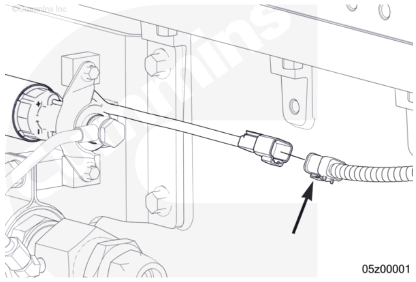


Remove the vibration isolators from each end of the fuel lift pump.



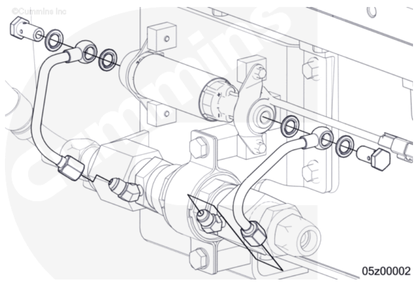
Note : Applications with lift pumps mounted on-engine are explained in the following steps.

Disconnect the electrical connection to the fuel lift pump.



⚠ WARNING ⚠

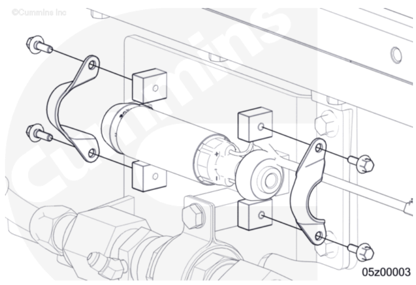
Depending on the circumstance, diesel fuel is flammable. When inspecting to performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, arcing equipment, or welding equipment) in the work area.



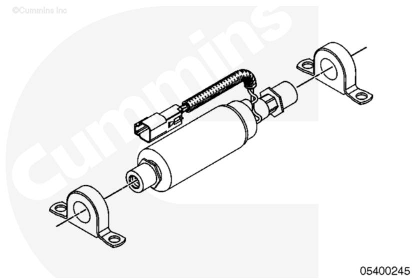
Remove the banjo capscrews and sealing washers and loosen the tube fittings securing the supply and discharge fuel lines from the lift pump.

Remove the fuel lines.

Remove the four capscrews securing the fuel lift pump to the mounting bracket.



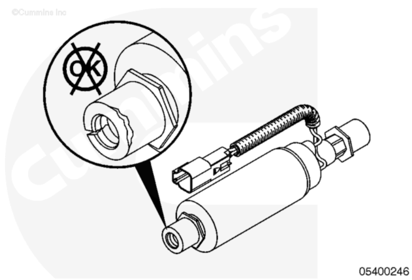
Remove the vibration isolators from each end of the fuel lift pump.



Inspect for Reuse

Inspect the fuel lift pump for any signs of damage.

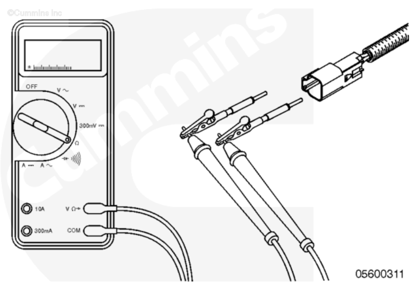
Replace the fuel lift pump, if damaged.



Use a digital multimeter to measure the resistance across the pins in the electrical connector. Reference the table below for resistance value. If **not** within specification, replace the fuel lift pump.

Fuel Lift Pump Resistance	
Minimum	**Maximum**
1 ohm	3 ohm

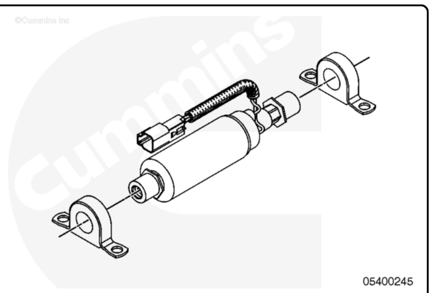
Use a digital multimeter. Check for a short between each pin and the fuel lift pump body. Replace the fuel lift pump, if shorted.



Install

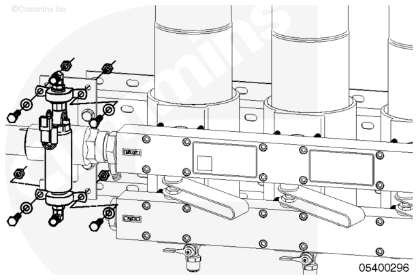
Note : Applications with lift pumps mounted on Stage 1 filter heads are explained in the following steps.

Install the vibration isolators at each end of the fuel lift pump.



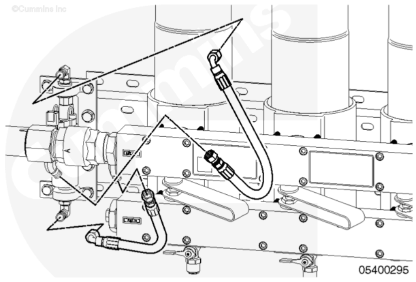
Install the fuel lift pump and four capscrews to the fuel filter mounting bracket

Torque Value: 15 n•m [133 in-lb]

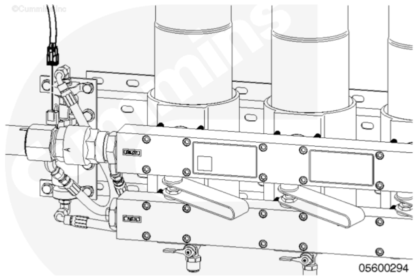


Connect the supply and discharge flexible hoses to the lift pump.

Torque Value: 54 n•m [40 ft-lb]



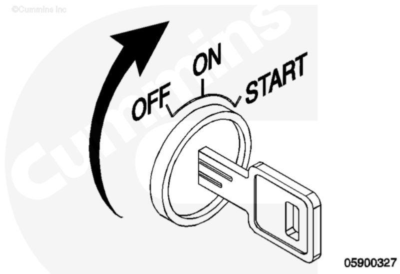
Connect the electrical connection to the fuel lift pump.



Turn the keyswitch to the ON position.

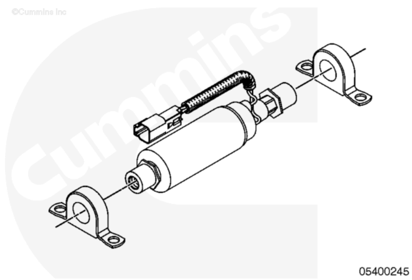
The fuel lift pump will operate for 120 seconds.

Check the fuel lift pump connections for leaks.



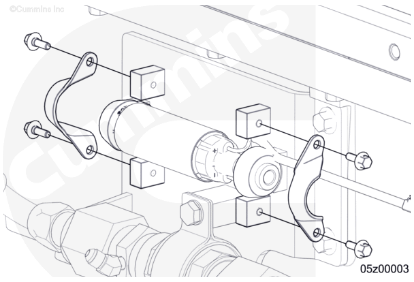
Note : Applications with lift pumps mounted on-engine are explained in the following steps.

Install the vibration isolators at each end of the fuel lift pump.



Install the electric fuel lift pump and four capscrews to the mounting bracket.

Torque Value: 15 n•m [133 in-lb]



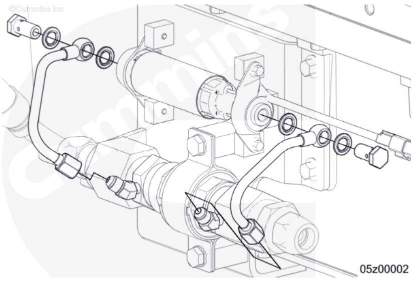
Install the banjo capscrews and new sealing washers securing the fuel supply and discharge lines to the lift pump.

Torque Value: 60 n•m [44 ft-lb]

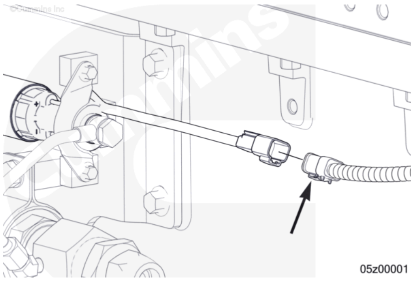
Rotate the tubes until the tube fittings are aligned with the 45 degree fittings on the check valve.

Tighten the fittings.

Torque Value: 20 n•m [177 in-lb]



Connect the electrical connection to the fuel lift pump.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the battery cables. See equipment manufacturer service information.
- Open the fuel supply valve.
- Turn the keyswitch to the ON position.
- The fuel lift pump will operate for 120 seconds.
- Check the fuel lift pump connections for leaks.

Last Modified: 10-May-2016